

Example 1

A beam is made of UNI EN 10083 25CrMo4 steel (yield strength $R_{eH} = S_y = 600$ MPa).

For each of the following three states of stress:

- $\sigma_{xx} = 0$ MPa (principal stress), $\sigma_{yy} = 280$ MPa, $\sigma_{zz} = 280$ MPa, $|\tau_{yz}| = 0$ MPa;
 - $\sigma_{xx} = -100$ MPa (principal stress), $\sigma_{yy} = 280$ MPa, $\sigma_{zz} = 280$ MPa, $|\tau_{yz}| = 0$ MPa;
 - $\sigma_{xx} = 0$ MPa (principal stress), $\sigma_{yy} = 280$ MPa, $\sigma_{zz} = 280$ MPa, $|\tau_{yz}| = 80$ MPa;
- compute the equivalent stress σ_{eq} for the failure criteria (Tresca and Von Mises);
 - use formulas to compute the safety factors F_s .
- [a] $F_s = 2.1$ b) $F_s = 1.6$ c) F_s (Tresca) = 1.7; F_s (Von Mises) = 1.9]

Example 2

A beam, made of UNI EN 10083 34CrNiMo6 steel (yield strength $R_{eH} = S_y = 800$ N/mm²) is loaded by a state of plane stress ($\sigma_{zz}=0$ MPa) in which the two principal stresses are proportional to each other ($\sigma_{xx}/\sigma_{yy} = -2.5$).

Draw the failure envelope for the Tresca failure criterion and compute the largest admissible stress σ_{xx} and σ_{yy} for a safety factor $F_s=1.5$.

$$[\sigma_{xx_max} = \pm 381 \text{ MPa}; \sigma_{yy_max} = \pm 152 \text{ MPa}]$$

Example 3

A structural component, made of a material whose yield strength is $R_e = S_y = 400$ N/mm², is loaded by a state of plane stress ($\sigma_{zz}=0$ MPa) with $\sigma_{yy} = 160$ MPa, $\sigma_{xx} = 0$ MPa. Compute the largest admissible value of the shear stress $|\tau_{xy}|$ for which there is no yielding according to a) the maximum shear stress criterion and b) the maximum distortion strain-energy criterion.

$$[a) \tau_{xy} = 183 \text{ MPa}; b) \tau_{xy} = 212 \text{ MPa}]$$

Example 4

Given the stress tensor at a point of a plate, made of a EN-GJL-350 grey cast iron (brittle material)

$$[\sigma] = \begin{bmatrix} \sigma_{xx} & \tau_{xy} & \tau_{xz} \\ \tau_{yx} & \sigma_{yy} & \tau_{yz} \\ \tau_{zx} & \tau_{zy} & \sigma_{zz} \end{bmatrix} = \begin{bmatrix} 180 & 30 & 0 \\ 30 & 55 & 0 \\ 0 & 0 & 0 \end{bmatrix} [MPa],$$

ultimate tensile strength $R_m = S_U = 350$ MPa, draw the failure envelope to evaluate, both graphically and analytically, the safety factor F_s . Comment on the suitability of the found value.

$$[F_s = 1.9]$$

Example 5

A brittle material is loaded by the stress components $\sigma_{xx} = -75$ MPa, $\sigma_{zz} = 135$ MPa, $|\tau_{xz}| = 60$ MPa, $\sigma_{yy} = 260$ MPa ($\tau_{xy} = \tau_{yz} = 0$ MPa). Compute the minimum value for the ultimate tensile strength of the material in order to avoid failure with a safety factor $F_S = 3$.

$$[S_U \geq 780 \text{ MPa}]$$

Example 6

A ductile material is loaded by the stress components $\sigma_{xx} = 110$ MPa, $\sigma_{yy} = -140$ MPa, $\sigma_{zz} = 0$ MPa, $\tau_{xy} = -150$ MPa, $\tau_{xz} = 0$, $\tau_{yz} = 0$ MPa. Compute the minimum value for the yield strength according to a) Tresca failure criterion and b) Von Mises failure criterion in order to avoid failure with a safety factor $F_S = 1.5$.

$$[\text{a) } R_{eH} = S_y = 585 \text{ MPa; b) } R_{eH} = S_y = 507 \text{ MPa}]$$

Example 7

Strains on the surface of a steel plate ($E=210 \cdot 10^3$ MPa, $\nu=0.3$, $R_{eH} = S_y = 355$ MPa), as measured by strain gages, are: $\varepsilon_{xx}=1000$ $\mu\text{m/m}$, $\varepsilon_{yy}=-600$ $\mu\text{m/m}$, $\gamma_{xy}= 450$ $\mu\text{m/m}$. Evaluate the safety factor F_S according to the Tresca failure criterion.

$$[F_S = 1.2]$$